Accelerating the Implementation of the ASEAN Sustainable Urbanisation Strategy – Phase II

Cities Inception Report

Phase II cities

This page intentionally left blank

TABLE OF CONTENTS

City infographics 1
City Technical Proposal design Methodology 4
Sihanoukville (Cambodia) – Solid Waste Management 6
Siem Reap (Cambodia) – Safety & Security 9
Pekanbaru (Indonesia) – Mobility 12
Makassar (Indonesia) – Urban Resilience 14
Oudomxay (Lao PDR) – Urban Resilience 16
Luang Prabang (Lao PDR) – Urban Resilience 18
Miri (Malaysia) – Urban Resilience 21
Iskandar Puteri (Malaysia) – Mobility 24
Yangon (Myanmar) – Safe Public Spaces 26
Cebu City (The Philippines) – Mobility 29
Caloocan City (The Philippines) – Housing 32
Nakhon Si Thammarat (Thailand) – Solid Waste Management 35
Chiang Mai (Thailand) – Safety & Security 37
Bac Giang (Vietnam) – Water, Waste & Sanitation 39
Hue (Vietnam) – Urban Resilience 41

CITY INFOGRAPHICS

Sihanoukville, Cambodia
Water, Waste & Sanitation

Proposed Direction
Improving Solid Waste Management

Scale of intervention
- City scale

Funding targeted
- Funding scoping is ongoing

1 Consultations — **9** Women — **22** Men

25 from the Government
2 from Private Sectors
2 from International Organizations
2 from CSOs

Siem Reap, Cambodia
Safety & Security

Proposed Direction
Improving Road safety, safe public spaces, digital safety and disaster risk management

Scale of intervention
- City scale

Funding targeted
- International Grant

2 Consultations — **6** Women — **32** Men

30 from the Government
5 from International Organizations
3 from Private Sector

Makassar, Indonesia
Urban Resilience

Proposed Direction
Improving Flood Management

Scale of intervention
- Metropolitan scale

Funding targeted
- Funding scoping is ongoing

2 Consultations — **8** Women — **16** Men — **2** Unstated

25 from the Government
1 from Academia

Pekanbaru, Indonesia
Mobility

Proposed Direction
Transport Planning & Active Mobility

Scale of intervention
- Metropolitan scale
- Neighbourhood scale

Funding targeted
- International Grant
- National budget

2 Consultations — **7** Women — **13** Men

18 from the Government
1 from CSOs
1 from Academia

Oudomxay, Lao PDR
Urban Resilience

Proposed Direction
Enhancing Drainage Systems

Scale of intervention
- Neighbourhood scale

Funding targeted
- Direct funding from international financing institution and National budget

1 Consultations — **4** Women — **16** Men

16 from the Government
4 from International Organizations

Luang Prabang, Lao PDR
Urban Resilience

Proposed Direction
Enhancing Drainage Systems

Scale of intervention
- Neighbourhood scale

Funding targeted
- Direct funding from international financing institution and National budget

1 Consultations — **3** Women — **18** Men

15 from the Government
4 from International Organizations
1 from Academia
1 from Media

Miri, Malaysia
Urban Resilience

Proposed Direction
Reducing coastal erosion, landslides and flooding

Scale of intervention
- Neighbourhood scale

Funding targeted
- International Grant
- Municipal budget

1 Consultations — 9 Women, 22 Men
25 from the Government
2 from Private Sector
2 from International Organizations
2 from CSOs

Iskandar Puteri, Malaysia
Mobility

Proposed Direction
Local revenue generation with Active Mobility

Scale of intervention
- Neighbourhood scale

Funding targeted
- Municipal budget

2 Consultations — 18 Women, 18 Men
28 from the Government
4 from Private Sector
3 from Academia
1 from CSOs

Yangon, Myanmar
Safety & Security

Proposed Direction
Safe Public Spaces

Scale of intervention
- Neighbourhood scale

Funding targeted
- Funding scoping is ongoing

1 Consultation — 6 Women, 23 Men
5 from International Organizations
22 from CSOs
2 from Private Sector

Caloocan, Philippines
Housing

Proposed Direction
Green Buildings & Public Spaces Improvement

Scale of intervention
- Neighbourhood scale

Funding targeted
- International Grant

2 Consultations — 10 Women, 16 Men
20 from the Government
4 from International Organizations
2 from CSOs

Cebu, Philippines
Mobility

Proposed Direction
Mobility Data Analytics & Command Centre

Scale of intervention
- Site scale

Funding targeted
- International Grant

2 Consultations — 12 Women, 21 Men
29 from the Government
4 from International Organizations

Nakhon Si Thammarat, Thailand
Water, Waste & Sanitation

Proposed Direction
Improving downstream Solid Waste Management

Scale of intervention
- Site scale

Funding targeted
- Provincial budget

3 Consultations — 27 Women, 25 Men
27 from the Government
16 individuals
5 from CSOs
3 from Academia
1 from Private Sector

Chiang Mai, Thailand
Safety & Security

Proposed Direction
Improving Safety & Security with digital tools

Scale of intervention
- Site scale

Funding targeted
- Provincial budget

3 Consultations — **10** Women — **27** Men

28 from the Government

5 from CSOs

2 from Academia

2 from Private Sector

Hue, Vietnam
Urban Resilience

Proposed Direction
Improving Solid Waste Management

Scale of intervention
- Neighbourhood scale

Funding targeted
- Direct funding from international financing institution

Point of attention
- National reform processes in Viet Nam have required more city-level coordination in the preparation of the Inception Report and Diagnostic Report, pushing them beyond the original delivery schedule.

2 Consultations — **8** Women — **16** Men

21 from the Government

3 from International Organizations

Bac Giang, Vietnam
Water, Waste & Sanitation

Proposed Direction
Improving drainage systems to separate wastewater with rainwater

Scale of intervention
- Site scale

Funding targeted
- Direct funding from international financing institution

Point of attention
- National reform processes in Viet Nam have required more city-level coordination in the preparation of the Inception Report and Diagnostic Report, pushing them beyond the original delivery schedule.

2 Consultations — **8** Women — **20** Men

19 from the Government

3 from International Organizations

1 from CSOs

5 Academia

Scale of intervention:

- *City: solutions implemented at a large scale across multiple neighbourhoods and sites*
- *Neighbourhood and Community: solutions implemented at a medium scale focusing on a single neighbourhood or community in a city*
- *Site and household: solutions implemented at a small scale, focusing on a single site, building or household.*

CITY TECHNICAL PROPOSAL DESIGN METHODOLOGY

How the City Technical Proposals are developed

The City Technical Proposals (CTPs), under the ASUS Project - Phase II project, are developed through a structured, multi-stage and participatory process ensuring that the proposed interventions are evidence-based, locally owned, and aligned with both city priorities, national strategies and with the Asean Sustainable Urbanisation Strategy.

Inception Phase

The process begins with the Inception Phase, which provides the foundation for sectoral analysis. During this stage, the project team works with city counterparts to map existing policies, plans, ongoing initiatives, and key stakeholders relevant to the selected priority sub-area. Stakeholder identification and mapping exercises help to clarify the institutional roles, coordination gaps, and opportunities for collaboration. The outputs of this phase are consolidated in the Inception Report, which provides an overview of the city context, existing frameworks, and initial hypotheses regarding key challenges and opportunities.

Diagnostic Phase

This is followed by the Diagnostic Phase, which deepens the analysis through desk research, data review, interviews, and multi-stakeholder workshops. Participatory tools such as Problem Tree analysis, SWOT analysis, and constraints and opportunity mapping are used to identify root causes, systemic barriers, and spatial dimensions of the priority issue. These exercises are conducted with technical departments, community representatives, civil society organisations, and, where possible, groups representing women, persons with disabilities, and other vulnerable populations. The results feed into the City Diagnostic Report, which provides a structured assessment of governance, finance, institutional capacity, spatial and sectoral conditions, and cross-cutting issues such as GEDSI and climate resilience. The Diagnostic Report concludes with strategic directions and recommendations that form the analytical basis for the CTP.

Co-design Phase

Building on this validated analysis, the process moves into the Co-Design Phase, where the focus shifts from "what is the problem?" to "what should be done about it?". In facilitated workshops, stakeholders engage in visioning exercises, scenario discussions, and solution tree development to identify feasible interventions. Technical, institutional, financial, and social dimensions of potential solutions are discussed, along with risks, implementation constraints, and opportunities for partnerships. This phase helps define the scope, components, and sequencing of the proposed intervention, while also clarifying roles and responsibilities among local actors.

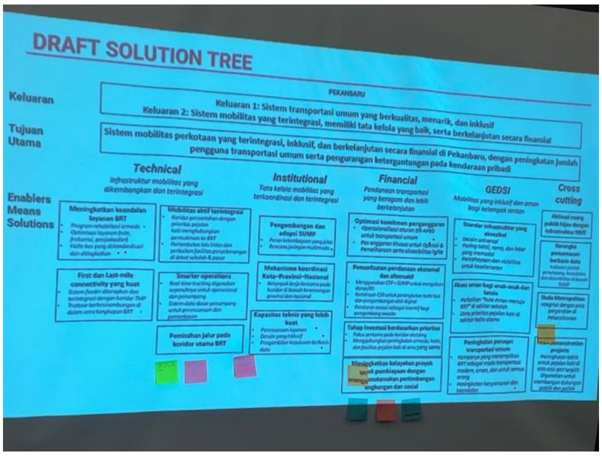

Figure 1 Solution tree exercise in a city workshop

City Technical Proposal Development

The insights from these consultations are then translated into the City Technical Proposal, where the diagnostic findings are used to develop a structured project concept. The CTP includes a clear problem statement and rationale, a Theory of Change, defined objectives and outcomes, key components and activities, governance and implementation arrangements, financing considerations, risk management, and a Monitoring, Evaluation and Learning (MEL) framework. Cross-cutting principles such as GEDSI, climate resilience, and inclusive governance are integrated throughout the proposal, ensuring that the intervention responds to differentiated needs and contributes to long-term institutional strengthening.

Throughout all phases, the process emphasises local ownership, iterative validation, and practical feasibility. Each key document — the Inception Report, Diagnostic Report, and City Technical Proposal — is developed in close collaboration with city authorities and validated through stakeholder consultations. This stepwise approach ensures that CTPs are not externally imposed project ideas, but locally grounded, technically sound proposals that can move toward feasibility, financing, and implementation.

Mainstreaming Gender Equality, Disability, and Social Inclusion (GEDSI)

GEDSI is systematically integrated across the ASUS Project - Phase II process and will be explicitly reflected in both the City Diagnostic Report and the City Technical Proposal, in line with the approved overall project design. Rather than being treated as a standalone topic, GEDSI considerations are embedded throughout the analytical, participatory, and planning stages.

Diagnostic report

During the diagnostic phase, GEDSI analysis is incorporated through stakeholder mapping and participatory tools that include representatives of women's groups, persons with disabilities, elderly people, youth, informal workers, and other marginalised urban communities when possible.

The Problem Tree and SWOT exercises explicitly examine how the core urban challenge and its root causes affect different social groups in distinct ways (e.g., safety, accessibility, affordability, exposure to climate or environmental risks). This informs the Social Inclusion dimension of the spatial and sectoral analysis and strengthens the problem statement by highlighting distributional impacts and barriers to access.

City Technical Proposal

In the co-design phase, GEDSI is further mainstreamed through visioning, scenario-building, and role-playing or empathy-based workshop exercises that help technical stakeholders understand lived experiences of different user groups (e.g., women travelling at night, persons with mobility or sensory impairments, low-income households). These exercises influence the prioritisation of solutions in the Solution Tree and early design of project components, ensuring that proposed interventions respond to differentiated needs rather than average users only.

Figure 2 Inclusive role-playing exercise during a city workshop

As a result, inclusivity is built into the selection of project sites, service standards, design features, and implementation approaches.

Within the City Technical Proposal, GEDSI is embedded under cross-cutting principles:

- design features that improve universal accessibility, safety, and affordability.
- institutional arrangements that ensure participation of under-represented groups in implementation and oversight; and
- indicators in the MEL framework that track who benefits from the intervention, with attention to gender, age, disability status, and socio-economic vulnerability

SIHANOUKVILLE (CAMBODIA) – SOLID WASTE MANAGEMENT

City Background

Sihanoukville, Cambodia's largest coastal city and deep-sea port, has undergone rapid urbanisation driven by tourism and industrial development. Visitor numbers rebounded to 5.16 million in 2023, while waste generation surged from about 100 tonnes per day in 2015 to 600 tonnes per day in 2025, with peaks of up to 900 tonnes in 2019.

Figure 3 Sihanoukville deep seaport area

Current resources and policies

Municipal solid waste collection has expanded, but coverage remains uneven, particularly in informal settlements and areas with narrow access roads. A UN-Habitat WaCT survey in 2021 found recycling rates at just 4%, with nearly all waste disposed at landfill. Waste is dominated by organics (44–49%) and plastics (23–31%), but sorting infrastructure is minimal. The municipality operates under concession agreements with private contractors, but irregular collection, weak monitoring, and low willingness to pay hinder system performance. The project is well aligned with Cambodia's policy framework, reflecting strong national and local commitment to sustainable solid waste management. It supports the National Policy on Solid Waste Management 2020–2030 (adopted 2021) and Sub-Decree No. 113 on Urban Solid Waste Management (2015), which promote decentralised municipal responsibility, improved service systems, and the 3Rs approach. It also aligns with the Land Management and Land Use Policy for Developing Preah Sihanouk Province 2022–2038 and the Smart City Action Plan for Preah Sihanouk Province (2024–2028) adopted in 2024, both of which prioritise modernised waste systems, recycling, and digital solutions. However, enforcement is inconsistent, and investment in recovery infrastructure is lacking the moment.

Priority challenges & opportunities

The city's challenges include unreliable collection, very low recycling rates, leachate leakage, and limited investment in waste recovery. Yet there are significant opportunities: large shares of organic and plastic waste could be diverted from landfill, and the tourism and SEZ-driven growth provides a strong incentive to modernise waste management.

Figure 4 Data collection exercise in Sihanoukville with city officials, operators, and CSOs

From the application form to the inception phase

The application form of the city identified Water, Waste and Sanitation as its main priority, proposing actions to strengthen smarter waste management, improve collection efficiency, promote public participation and recycling, and apply digital solutions. During the inception phase, site visits and consultations highlighted specific gaps: irregular collection in informal/remote areas, heavy reliance on landfill, weak enforcement, low public awareness and limited data and fee-tracking systems. As a consequence, the technical direction Integrated digital systems for customer registration, fee collection, and service monitoring. The technical assistance to the city is focusing on:

- Law enforcement and public outreach to improve segregation and payment compliance.
- Support to the informal recycling chain (e.g. waste pickers, TonToTon) to increase recovery and safety.
- A strategic plan to cut landfill waste, including composting feasibility and plastic-reduction measures.

Next steps & planned support

Theproject teamwill project team support the city in developing a City Technical Proposal focusing on the improvement of waste collection service management efficiency, governance support with law enforcement and public awareness campaign, strengthening the existing informal value chain of recycling activities, developing a plan to reduce landfill waste, and improving collection services in underserved areas::

- **Q4 2025**: Completion of the city diagnostic workshop
- **Q1 2026**: additional data collection consultations for the City Technical Proposal development
- **June 2026**: Completion of the City Technical Proposal

Figure 5 Problem tree exercise in Sihanoukville with city officials, operators and CSOs

Potential funding sources

- Grant Application from Korea Smart City Cooperation Fund

ASUS Working Group composition

City focal point	H.E. Pheap Em, Deputy Governor of Preah Sihanouk Province
Steering committee	**Sihanoukville Smart City Working Group** **Chair** Mr. Em Pheap, Deputy Governor of Preah Sihanouk Province **Vice chair** Mr. Sar Kakada, Mayor of Sihanoukville Mr. Cheng Srong, Director of Provincial Department of Land Management, Urban Planning, Construction and Cadaster Members Mr. Thai Rithy, Deputy General Director of Sihanoukville Autonomous Port Mr. Chhet Ratanak, Deputy Director of Provincial Hall Administration Mr. Chen Ren, Deputy Director of Provincial Hall Administration Mr. General Nop Panha, Deputy Commissioner of Provincial Police Commissioner Mr. Chey Vimean Rith, Director of Provincial Department of Economic and Finance Mr. Samuth Sothearith, Director of Provincial Department of Environment Mr. Borey Vongsanith, Director of Provincial Department of Public Work and Transport

	Mr. Yuos Sothea, Director of Provincial Department of Post and Telecommunication Ms. Lim Samarn, Director of Provincial Department of Health Mr. Tor Mony Sak, Director of Provincial Department of Planning Taing Sochetkrisna, Director of Provincial Department of Tourism Mr. Nen Chamroeun, Director of Provincial Department of Agriculture, Forestry and Fisheries Mrs. Sao Kan, Director of Provincial of Women Affairs Mr. Heng Sohonrith, Director of Provincial Department of Water Resource and Meteorology Mr. Soam Savath, Director of Provincial Department of Industry, Science, Technology and Innovation Mr. Prak Somphrathna, Director of Provincial Department of Mine and Energy Mr. Soa Lim, Director of Electricity of Preah Sihanouk Province Mr. Hang Phirak, Deputy Director of Provincial Department of Land Management, Urban Planning, Construction and Cadaster Ms. Noun Sopheap, Director of Financial Unit of Provincial Hall Mr. Kheang Phearum, Director of Planning and Investment of Provincial Hall Mr. Prak Visal, Director of Public Relation and International Cooperation of Provincial Hall, Member Mr. Bou Voatna, Director of Inter-sectoral Unit of Provincial Hall

	Mr. Heb Sokhannaro, Director of Landfill Mr. Ly Chetniyum, Chief of Development and Construction Management Office Mrs. Nai Saly, Chief of International Relation Office of Provincial Hall Mr. Vannak SopheaNuth, Chief of Investment Office of Provincial Hall Mr. Tieng Vannarith, Director of Administration of Preah Sihanouk Municipality Mr. Kim Thaisor, Chief of Land Management, Urban Planning, and Construction Office of Preah Sihanouk Municipality Mr. Chhit Sophat, Chief of Sangkat 1 Ms. Samreth Sokhemera, Chief of Sangkat 2 Mr. Kan Leoung, Chief of Sangkat 3 Mr. Seng Nim, Chief of Sangkat 4 Representatives of other development partners Representative from the Ministry of Interior (ASCN)

Figure 6 Workshop in Sihanoukville

SIEM REAP (CAMBODIA) – SAFETY & SECURITY

City Background

Siem Reap, Cambodia's tourism capital and gateway to Angkor, has a population of about 273,000 residents and received over 3.2 million visitors in 2024, with projections of 11 million by 2035. Tourism is the backbone of the local economy, making urban safety, cleanliness, and public space management critical to sustaining the city's brand and visitor experience.

Figure 7 Stakeholder consultation in Siem Reap

Current resources and policies

Road safety is a pressing issue: in 2024, the city recorded 131 traffic crashes, resulting in 110 deaths and 180 injuries. Local authorities have established traffic enforcement units, but weak coordination, poor junction design, and limited signage undermine effectiveness.

The project is grounded in Cambodia's established policy framework on urban safety and security, providing a strong baseline for sustained action and future advocacy. It aligns with the Pentagonal Strategy Phase I (2023), which prioritises safer communities, traffic safety, and disaster risk management, and with the Safe Village–Commune Policy (revised 2021), which promotes local-level crime prevention and public order. In the transport sector, the Law on Road Traffic (2015) and the National Road Safety Policy (2017) provide the legal and strategic basis for improving road safety, enforcement, and public awareness. At the city level, the Tourism Development Master Plan Siem Reap (2021–2035) and the Land Use Master Plan of Siem Reap City 2035 integrate safety, emergency response, and smart technologies into urban and tourism planning. These are further reinforced by the Siem Reap Smart City Roadmap (endorsed 2023), which identifies Smart Safety and Security as a priority sector. Together, these policies demonstrate clear institutional commitment and highlight strategic entry points for strengthening enforcement, smart systems, and community-based safety initiatives in the long term.

That said, but implementation remains fragmented across police, transport, and city departments. Public space cleanliness is supported by municipal cleaning services, yet illegal dumping persists and affects the city's image. Early warning systems exist in basic form, such as lightning warnings for visitors in Angkor Park, but these remain limited and underdeveloped.

Priority challenges & opportunities

Siem Reap faces important challenges around traffic safety, incident reporting, and the cleanliness of public spaces. The tourism rebound has increased these issues, as more visitors mean more pressure on roads, services, and city management. At the same time, there are clear opportunities: the provincial government has a mandate to expand parks and green areas, and improvements in public realm management could directly support both resident well-being and tourism competitiveness.

Figure 8 Mapping exercise and SWOT analysis related to urban safety in Siem Reap

From the application form to the inception phase

The city selected Water, Waste, Sanitation and Safety & Security as its two main priorities to focus on with the ASUS Project - Phase II. The city proposed smart city measures such as smart waste management, IoT-based monitoring, CCTV and early-warning systems. Through the inception phase, fieldwork and stakeholder consultation highlighted that safety and security require urgent, targeted action.

Figure 9 Problem tree exercise in Siem Reap

Identified issues were fragmented CCTV networks lacking integration, rising road accidents linked to weak traffic design and enforcement, limited reporting mechanisms, a shortage of safe public spaces, and vulnerability to climate hazards such as storms and lightning. The technical assistance was therefore refined to focus on safety & security, considering the multi-dimensional nature of the priority sub-area, looking at road safety, safe public spaces, digital safety and disaster risk management.

Next steps & planned support

The project team will support the city in developing a City Technical Proposal that prioritises safety upgrades, particularly at high-risk intersections, alongside measures to improve public space cleanliness and management. Proposed interventions will address GEDSI issues related to urban safety and security. Under the City Technical Proposal, Training will be proposed on crash data management, enforcement coordination, and public communications targeting both residents and visitors:

- **Q4 2025**: Completion of the city diagnostic workshop
- **Q1 2026**: additional data collection consultations for the City Technical Proposal development
- **June 2026**: Completion of the City Technical Proposal

Figure 10 Problem tree validation with city officials, CSOs and development partners in Siem Reap

Potential funding sources

- Grant Application from Korea Smart City Cooperation Fund

ASUS Working Group composition

City focal point	H.E. Linne Yun, Deputy Governor, Siem Reap Province
Steering committee	**Technical Working Group for Smart Security and Safety** **Chair** H.E Prak Sophoan, Governor of Siem Reap Province **Vice-Chair** Mr. Yun Linne, Deputy Governor, Siem Reap Province **Members** Mr. Samkol Sochetra, Deputy Governor, Siem Reap Province Mr. Ly Vannak, Director of Administration, Siem Reap Province Mr. Huot Sothy, Commissioner of the Provincial Police Mr. Ky Virin, Director of Department of Public Works and Transport

	Mr. Tip Piseth, Head of Planning and Investment Unit, Siem Reap Province Ms. Lim Phallika, Deputy Governor of Siem Reap Municipality Mr. Ek Cheachamreoun, Head of Siem Reap City Police Department Mr. Hea Hao, Chief Sangkat Svay Dangkom Mr. Takeharu Mizukoshi, Director of Forval Company

PEKANBARU (INDONESIA) – MOBILITY

City Background

Pekanbaru, the capital of Riau Province, is one of the fastest-growing cities in Sumatra with a population of about 1.12 million. Strategically located on the Siak River, it has developed as a metropolitan anchor with strong trade, logistics, and industrial functions. Mobility has been a key focus, with a decade-long effort to develop a bus-based mass transit system to manage urban growth and congestion.

Figure 11 Site visit of Pekanbaru's bus terminal

Current resources and policies

The Trans Metro Pekanbaru (TMP) system currently covers 10 corridors with 110 buses, although only about 30 percent of the fleet is operational. Plans exist to expand with 12 feeder corridors. Since March 2025, students ride free, with over 5,500 registered users and about 400 daily riders. Real-time bus tracking is being integrated into the national MitraDarat application. The city collaborates with GIZ through SUTRI NAMA–Indobus and with the Institute for Transportation and Development and Policy (ITDP), which is conducting a study on electrification of the bus fleet.

Priority challenges & opportunities

TMP faces multiple challenges: an under-utilised fleet as 70% of it requires major repair, reliability issues, and uncertainty around financing for feeder services. Nevertheless, policy backing is strong, and significant opportunities exist to expand coverage, improve service quality, and transition towards low-emission mobility. Electrification of the bus fleet offers long-term cost and environmental benefits, while feeder services could expand accessibility across the city.

From the application form to the inception phase

Pekanbaru's application form (Jan 2025) set mobility as its priority, proposing broad actions to expand and upgrade TMP Bus Rapid Transit system with feeder services, improve bus stops and pedestrian links, add e-ticketing and other smart tools, and integrate cycling and walking facilities**.** During the inception phase (May 2025), consultations and field visits revealed key gaps—very low ridership, ageing buses, weak last-mile connections, overlapping city–province road authority, and the absence of a Sustainable Urban Mobility Plan (SUMP). The technical assistance was therefore narrowed to focus on preparing a SUMP as the city's integrated mobility framework, improving first and last-mile connectivity with active mobility (walking/cycling).

Next steps & planned support

The project team will support the city in developing a City Technical Proposal that sets out a SUMP for Pekanbaru. The proposal will identify priority non-motorised transport projects to improve accessibility to TMP, outline a phased plan for service expansion, and incorporate an electrification strategy. As part of the activities proposed under the City Technical Proposal, capacity building will focus on sustainable mobility planning, performance monitoring, and integration of Intelligent Transport Systems for better service quality and passenger information:

- **Q4 2025**: Completion of the city diagnostic workshop to complete the diagnostic report, identifying GEDSI issues.

- **February 2026**: co-design workshop for the City Technical Proposal development, validating solutions addressing GEDSI root causes that are impacting the transport system performance.
- **June 2026**: Completion of the City Technical Proposal

Potential funding sources

- Grant Application from City Climate Gap Fund
- National Budget Programming

Figure 12 Mr, Sunarko, Head Department of Transport of Pekanbaru

ASUS Working Group composition

City focal point	Mr. Agung Nugroho, Mayor of Pekanbaru
Steering committee	**City Focal Point (Technical Staff):** • Mr. Sunarko, Head Department of Transport. • Ms. Solid Bunda, Department of Transport • Mr. Todi Kurniawan, Planning and Development Agency
	Steering Committee: • Mr. Alek Kurniawan Head of Planning and Development Agency • Ms. Nadya, Department of Environment • Mr. Dedi Harianto Planning and Development Agency (Riau Province) • Mr. Yulismor, Department of Transport (Riau Province) • Ms. Fivi Zulfianingsih, Department of Public Works, Spatial Planning, Housing, Human Settlements, and Land Tenure (Riau Province). • Mr. Dail Umamil Asri, Director of Connectivity and Logistic Infrastructure, Ministry of National Planning and Development/Bappenas (ACSI representative) • Mr. Gensly Bachtiar, Ministry of Home Affairs (ASCN representative)

Figure 13 Site visit of BRT stations to assess accessibility level

MAKASSAR (INDONESIA) – URBAN RESILIENCE

City Background

Makassar, the economic hub of eastern Indonesia, is a rapidly growing coastal metropolis highly exposed to climate-related hazards. Flooding is the most frequent and damaging risk, with severe events recorded in December 2024 and February 2025 displacing more than 2,000 people and damaging hundreds of homes. The city's topography, rapid urbanisation, and intensifying rainfall patterns have made flood management a central concern for city planners and residents alike.

Current resources and policies

Makassar has already laid an important institutional foundation for disaster risk reduction (DRR). The city has enacted DRR studies and plans with legal backing, supported by both a city regulation and a mayoral decree. Ongoing structural measures include the Nipa-Nipa regulatory pond and drainage rehabilitation projects. At the community level, settlement-scale nature-based solutions have been piloted through the RISE programme, which introduced rain gardens and bio-filtration systems in vulnerable neighbourhoods.

Figure 14 Inception workshop in Makassar

Priority challenges & opportunities

Makassar's main challenges stem from low-lying topography, insufficient drainage, and growing pressure on land from rapid population growth. Climate change is further intensifying these risks, leading to more frequent and severe flood events. On the other hand, the city has a strong enabling environment: clear institutional mandates, political support, and a pipeline of both structural and nature-based measures that can be scaled up. Universities and local research institutions have also been actively engaged, providing a strong knowledge base for future resilience initiatives.

From the application form to the inception phase

Makassar's application form highlighted urban resilience as its key priority, broadly aiming to address flooding, sea-level rise, and extreme weather through stronger infrastructure, early-warning systems, and disaster-preparedness plans. During the inception phase (May 2025), fieldwork and stakeholder consultations revealed critical gaps—recurrent severe floods, ageing and undersized drainage, overlapping river-management authority across city, provincial, and national agencies, rapid land conversion in neighbouring regencies, and low public awareness on waste and flood preparedness. The project will therefore focus to a concrete flood-risk reduction strategy, combining structural and nature-based solutions (e.g., drainage upgrades, Nipa-Nipa regulatory pond), strengthening institutional capacity and cross-jurisdiction coordination under the Mamminasata metropolitan framework, aligning with national and city plans (RPJMD), mobilising external finance, and boosting community engagement.

Next steps & planned support

The project team will support the city in developing a City Technical Proposal that prioritises interventions in flood-prone sub-catchments and outlines the optimisation of the Nipa-Nipa pond, integrated planning across sectors, including infrastructure upgrades and nature-based solutions to address recurrent flooding, and improved governance to address the overlapping authority river management. The project team will also provide capacity-building for city agencies and communities on flood incident management, risk-informed operations and maintenance, and the upkeep of nature-based solutions. This combination of institutional strengthening and targeted investments will help Makassar reduce the impacts of future floods:

- **Q4 2025**: Completion of the city diagnostic workshop
- **Q1 2026**: co-design workshop for the City Technical Proposal development
- **June 2026**: Completion of the City Technical Proposal

Figure 15 Site visit of flood-prone areas in Makassar with response team and city officials

Potential funding sources

The funding sources for implementing the City Technical Proposal are still being scoped at this stage.

ASUS Working Group composition

City focal point	Mr. Munafri Arifuddin, Mayor of Makassar
Steering committee	**City Focal Point (Technical Staff):** • Ms. Nur Ismawaty, Department of Communication and Information; • Mr. Zainuddin Rani, Planning and Development Agency **Steering Committee**: • Mr. Fuad Azis, Department of Spatial Planning, • Mr. Ismunandar, Local Disaster Risk Reduction Agency (City of Makassar) • Mr. Ronny Narra, Department of Public Works • Mr. Ansuar, Department of Housing and Human Settlements • Mr. Restu Ramadhan, District of Manggala. • Mr. Gensly Bachtiar, Ministry of Home Affair (ASCN representative) • Mr. Dail Asri, Bappenas (ACSI representative)

OUDOMXAY (LAO PDR) – URBAN RESILIENCE

City Background

Oudomxay City, the capital of Oudomxay Province in northern Laos, is home to about 93,000 residents across 97 villages. Its strategic location makes it a transit node linking Luang Prabang, China, and other parts of northern Laos. The city has grown rapidly, but mountainous terrain and river systems constrain development and expose the population to recurrent flooding and environmental pressures.

Figure 16 Inception workshop in Oudomxay with city officials

Current resources and policies

Several resilience-related programmes are underway. UNEP and GEF have supported drainage improvements that form the basis of the planned ADB City Development Project II, while the World Bank has financed flood protection works, including gates and embankments. A City Master Plan from 2013 identified settlement upgrading, flood control, and road infrastructure as priorities. Ecosystem-based adaptation approaches are increasingly being mainstreamed into city planning.

The proposed direction is closely aligned with Lao PDR's national and local policy framework, providing a strong institutional baseline and entry points for long-term advocacy. At the city level, the Xay City Master Plan (2013) sets the spatial and infrastructure development direction, including drainage, road upgrading, and settlement reorganisation. Nationally, the 9th National Socio-Economic Development Plan (2021–2025) prioritises climate resilience, sustainable urban infrastructure, and environmental management, directly supporting investments in flood control, waste systems, and resilient urban services. The project is also consistent with the National Green Growth Strategy of Lao PDR (2019), which promotes environmentally sustainable urban development and resource-efficient infrastructure. In the environmental governance domain, the Environmental Protection Law (revised 2012, amended 2019) and associated ESMP requirements provide the regulatory basis for integrating safeguards, watershed management, and pollution control into infrastructure planning. In addition, Lao PDR's National Urban Development Strategy (2012) supports balanced urban growth, improved basic services, and stronger local planning systems. These policies demonstrate a clear national commitment to resilient and sustainable urban development, and they provide a strong policy foundation to sustain and scale the project outcomes beyond the potential implementation period.

Priority challenges & opportunities

The main challenges for Oudomxay City lie in frequent flooding, incomplete drainage systems, inadequate solid waste services, and limited municipal capacity to manage rapid urbanisation. However, the city benefits from an unusually high level of donor engagement, creating opportunities to leverage and coordinate multiple investments. There is strong potential to stitch together drainage, flood protection, waste management, and road upgrades into a coherent resilience package that maximises impact and sustainability.

From the application form to the inception phase

The application form of Oudomxay City (January 2025) identified urban resilience as its top priority, citing rapid population growth, informal settlement expansion, weak drainage, and rising flood risks. The city proposed broad solutions: strengthening urban planning and construction regulation, improving waste and sewage management, and building a sustainable urban planning management system, supported by local budgets, national transfers, and international aid.

Figure 17 Inception workshop in Oudomxay with city officials

During the inception phase (April 2025), field visits and stakeholder consultations highlighted more specific needs, including insufficient drainage systems and lack of water gates, which lead to flooding during the rainy season, as well as urban expansion extending beyond the City Master Plan. These findings pointed to critical gaps in climate-resilient infrastructure and urban planning. The project will therefore focus on improving flood management and constructing drainage systems as the core of the project in the city.

Next steps & planned support

The project team will work with the city to develop a City Technical Proposal that will focus on reducing the impact of flooding with nature-based solutions and improving drainage systems to enhance climate resilience and support the City's Master Plan. Means of implementation will address GEDSI root causes of flooding. For the implementation of the City Technical Proposal, two directions are explored with city authorities: the National budget programming and international fund sources:

- **Q4 2025**: Completion of the city diagnostic workshop and data collection consultations for the City Technical Proposal development
- **May 2026**: Completion of the City Technical Proposal

Figure 18 Inception workshop in Oudomxay with city officials

Potential funding sources

- National Budget programming
- Other international funding sources

ASUS Working Group composition

City focal point	Mr. Anousack Vongphachanh, Technical Staff, Housing and Urban Planning Sector
Steering committee	Representatives of: • Provincial Governor's Office (Oudomxay Province) • Xay City Administration Office • City Development and Administration Organisation • Provincial Department of Public Works and Transport • District Department of Public Works and Transport • Provincial Department of Agriculture and Environment • District Department of Agriculture and Environment • Office of Planning • Department of Housing and Urban Planning, Ministry of Public Works and Transport (ASCN representative) • Department of Planning and Finance, Ministry of Public Works and Transport (ACSI representative)

LUANG PRABANG (LAO PDR) – URBAN RESILIENCE

City Background

Luang Prabang, a UNESCO World Heritage city, remains one of the most visited tourist destinations in Laos. Its rapid growth has outpaced basic water, waste, and sanitation services, putting both public health and the city's global tourism brand at risk. Despite significant donor engagement, the scale and complexity of demand continue to outstrip current capacity, leaving drainage, solid waste, and wastewater management systems under pressure.

Figure 19 Bilateral meeting with Luang Prabang's stakeholders during the ASUS national event in Vientiane (Lao PDR)

Current resources and policies

Several major programmes are underway. ADB is investing USD 45 million in urban environment improvement, focusing on solid waste and drainage upgrades. The project for Improvement of Management Capacity of Water Supply Sector (JICA's MaWaSU3) from JICA is strengthening capacity in water supply services, while GRET has supported wetland preservation, now institutionalised within DPWT. Despite these efforts, gaps remain in coverage, operations, and financial sustainability. Local policies exist, but coordination and human resource capacity are uneven, limiting the effectiveness of these interventions.

The proposed project aligns with the 9th National Socio-Economic Development Plan (2021-2025) and the 9-year Water Supply and Sanitation Sector Development Strategy (2022-2030), which prioritise resilient urban infrastructure and improved water and sanitation services. The enhancement of water sector, including the flood management, is also consistent with the 9th National Socio-Economic Development Plan (2021–2025) prioritises and the 5-year development plan of Luang Prabang province (2021-2025), and Luang Prabang Smart and Integrated Urban Strategy.

By strengthening the capacity of Department of Water Supply (DWS) and Nam Papa State-owned Companies (NPSEs) at the provincial level, the project is fully integrated into ongoing regulatory reforms; capacity of planning, operation, and management of water sector; capacity of enforcement the regulations and monitoring the progress; capacity of planning the water supply system maintenance; and strengthening the corporation between DWS and NPSEs. Together, these initiatives create a strong foundation to sustain and water and sanitation services in Luang Prabang beyond the project period.

Priority challenges & opportunities

Luang Prabang faces persistent challenges: limited wastewater treatment facilities, low recycling rates, and weak public awareness of sanitation practices. Coordination among agencies is fragmented, and cost-recovery mechanisms remain inadequate. Yet the city has a strong opportunity base: a robust pipeline of donor-funded projects, alignment with ASEAN Smart Cities Network goals, and an urgent need to sequence investments in ways that build system resilience.

From the application form to the inception phase

Luang Prabang's application form (January 2025) highlighted a broad agenda across housing, mobility, and water, waste, and sanitation, aiming to manage urban sprawl, improve traffic management, and modernise waste services while preserving its UNESCO heritage. During the inception phase (May 2025), field visits and stakeholder consultations identified urban resilience—especially the ability to cope with climate risks, floods, and pressure from rapid tourism growth—as the most urgent and cross-cutting challenge. Key gaps included limited flood-resilient infrastucture, fragmented planning across departments, insufficient data and monitoring systems, and low community preparedness. The project will therefore focus on Urban Resilience, centering on:

- Integrated flood management through the construction of bridges and the upgrading of gravel roads, incorporating nature-based solutions and climate-resilient infrastructure standards.
- Strengthen institutional capacity and inter-agency coordination to plan, finance, and implement resilience measures.
- Community awareness and engagement to improve risk preparedness and support participatory urban planning.
- Alignment with national urban development strategies to unlock finance and sustain long-term resilience.

Figure 20 Opening remarks during the bilateral meeting between UN-Habitat Lao PDR and Luang Prabang's city officials (Lao PDR)

Next steps & planned support

The project team will work with the city to develop a City Technical Proposal that will focus on strengthening urban resilience through flood management, specifically the construction of elevated bridges and the upgrading of red soil roads into gravel roads to ensure year-round accessibility. GEDSI considerations will be incorporated in the City Technical Proposal. It will also emphasise enhancing increasing institutional capacity and community preparedness to better manage climate risks. For the implementation of the City Technical Proposal, two directions are being explored with city authorities: national budget programming and international funding sources:

- **Q4 2025**: Completion of the city diagnostic workshop and data collection consultations for the City Technical Proposal development
- **May 2026**: Completion of the City Technical Proposal

Potential funding sources:

- National Budget programming
- Other international funding sources

ASUS Working Group composition

City focal point	1. Mr. Yengher Vacha, Deputy Head of City Administrative Office, Coordinator of International Cooperation 2. Mr. Vinath Mekthanavanh, Vice Dean of the Faculty of Engineering, Souphanouvong University 3. Mr. Thanouphet Kittivong, Technical Officer, Housing and Urban Planning and Water Supply Sector
Steering committee	• Representatives of the Provincial Governor's Office (Luang Prabang Province) • Representatives of Luang Prabang City Administrative Office • Representatives of City Management and Services Office • Representatives of Provincial Department of Public Works and Transport • Representatives of District Department of Public Works and Transport

	• Representatives of Provincial Department of Agriculture and Environment • Representatives of District Department of Agriculture and Environment • Representatives of Office of Planning • Department of Housing and Urban Planning, Ministry of Public Works and Transport (ASCN representative) • Department of Planning and Finance, Ministry of Public Works and Transport (ACSI representative)

MIRI (MALAYSIA) – URBAN RESILIENCE

City Background

Miri, Sarawak's second city and Malaysia's first oil town, is branded as the "Resort City." It is a coastal and riverine urban centre with more than 94% of its population living in urban areas. Miri has faced intensifying climate hazards: in January 2025, its worst flood in a decade displaced nearly 6,000 residents in Sarawak State, while August 2024 brought a severe heatwave. Coastal erosion at Lutong Beach has also required temporary protection works.

Figure 21 Inception workshop in Miri (Malaysia)

Current resources and policies

Miri's economy remains diversified, combining petroleum with services, tourism, and logistics. The city is managed by the Miri City Council, which is increasingly confronted with the task of balancing economic development with hazard mitigation. Protected areas such as Mulu and Niah National Parks are important assets for eco-tourism but also vulnerable to climate stress.

Miri's efforts to mainstream urban resilience in their city development are supported by several policy frameworks. They serve as a basis for climate resilience and sustainable urban development. At the national level, these efforts are supported by the 12th Malaysia Plan (2021–2025), which prioritises climate resilience, disaster risk reduction, and low-carbon urban growth, as well as the National Policy on Climate Change (2009) and the National Disaster Management Policy (2018), which guide climate adaptation and coordinated disaster risk governance. In environmental planning, the Town and Country Planning Act (1976) and the Environmental Quality Act (1974) establish the legal basis for risk-sensitive land-use planning and environmental safeguards. At the state level, the Miri's efforts are aligned with the Sarawak State Planning Framework and the Sarawak Transport Master Plan 2025–2040, which promote resilient infrastructure and integrated urban systems. Locally, Miri's smart-city and resilience initiatives, including the Kenyalang Smart City Development framework (ongoing), provide an operational platform to embed climate adaptation into service delivery. These policies demonstrate clear multi-level commitment and offer strategic entry points for the readiness of a City Technical Proposal focusing on urban resilience at the institutional level.

Priority challenges & opportunities

Miri's vulnerabilities are multifaceted: flooding, landslides, coastal erosion, and heatwaves all pose risks to residents and economic assets. Yet the city also has strong opportunities to integrate resilience measures into its identity as a resort and eco-tourism hub. There is an opportunity to bundle coastal protection, drainage improvements, and early-warning systems together, positioning Miri as a secondary city model for climate adaptation measures.

Figure 22 Site visit of targeted areas prone to coastal erosion in Miri (Malaysia)

From the application form to the inception phase

Miri City's application form (January 2025) highlighted a broad vision to become a Green, Smart, and Most Liveable International Resort City by 2030, focusing on green initiatives, smart service delivery, and social inclusion (children, elderly, and rural-urban migrants). It listed multiple solutions, including smart city applications (Miri CARES, Smart Bus/Truck, Smart Drain), green infrastructure, renewable waste management, and stronger community programs.

During the inception phase (July 2025), detailed field visits and stakeholder consultations revealed that urban resilience is the most urgent and cross-cutting challenge, with Miri facing severe climate-related risks—flash floods, landslides, coastal erosion, heatwaves, and bushfires—combined with ageing drainage and waste infrastructure, jurisdictional gaps, and limited technical capacity.

The project will therefore focus on urban resilience, aiming to integrate smart city tools (e.g., smart drainage, Miri Smart City Command Centre) into a city-wide climate adaptation strategy. Planned actions include integrated flood and slope management, coastal protection, nature-based solutions, improved waste and leachate treatment, stronger inter-agency coordination, and expanded community engagement and capacity training.

Next steps & planned support

The project team will support the city in developing a City Technical Proposal to mitigate landslides, coastal erosion, heatwaves and flooding caused by illegal dumping and the transition to landfilling. The City Technical Proposal will build upon existing initiatives by:

- **Enhancing Data-Driven Decision-Making**: Integrating data from smart systems into a unified platform can improve forecasting and response strategies for urban resilience challenges.
- **Expanding Community Engagement**: Leveraging platforms like Miri CARES to involve residents in resilience planning and implementation ensures that solutions are community-driven and context-specific.
- **Promoting Sustainable Infrastructure**: Aligning with the Kenyalang Smart City development, the ASUS Project - Phase II can support the adoption of green technologies and sustainable practices in urban planning and development.
- **Strengthening Governance and Collaboration**: Facilitating partnerships between local authorities, residents, and stakeholders can enhance the effectiveness of resilience initiatives and ensure long-term sustainability.

Activities planned in 2026:

- **Q1 2026**: Co-design workshop for the City Technical Proposal development
- **June 2026**: Completion of the City Technical Proposal

Figure 23 Site visit of targeted areas prone to landslides in Miri (Malaysia)

Potential funding sources

- Municipal budget
- State Government budget
- International budget

ASUS Working Group composition

City focal point	Mdm. Siti Suhana bte Rosli (C.A.M), City Treasurer & Chief Information Officer
Technical working group	• Mr. Abdul Rahman Bin Taupek, Ag. City Secretary - LEAD

	• Mr. George Anak Gumbang, Assistant Secretary • Mdm. Jenifer Leku Balang, Transformation & Innovation Section • Mdm. Siti Suhana bte Rosli (C.A.M), City Treasurer & Chief Information Officer • Mdm. Dayang Siti Nurbaya Binti Kipli, Assistant Secretary - Corporate Section • Mdm. Marsha Bathsheba Anak Moulton King, Assistant Legal Officer

ISKANDAR PUTERI (MALAYSIA) – MOBILITY

City Background

Iskandar Puteri, in Johor, Malaysia, is a flagship city within the Johor–Singapore Special Economic Zone (JS-SEZ). With nearly 581,000 residents and a GDP of over USD 10.5 billion in 2023, it is a fast-growing urban centre with key districts including Medini, Puteri Harbour, Kota Iskandar, and Port Tanjung Pelepas.

Figure 24 Site visit in Iskandar Puteri (Malaysia)

Current resources and policies

The city is well-served by a bus network and is positioned to benefit from regional rail projects such as the Rapid Transit System (RTS) and planned High-Speed Rail (HSR) and Light Rail Transit (LRT) lines. Bukit Indah functions as a major circulation node with strong residential, retail and recreation pull, but wayfinding remains car-oriented, and public-realm continuity between hubs is weak. Governance is under the Majlis Bandaraya Iskandar Puteri (MBIP),, supported by the Iskandar Regional Development Authority (IRDA) at the regional level.

Figure 25 Parking area of commercial neighbourhood in Iskandar Puteri (Malaysia)

Priority challenges & opportunities

Iskandar Puteri's challenge lies in ensuring first- and last-mile connectivity, overcoming car dependency, and improving pedestrian and cycling infrastructure. Yet the city has major opportunities to prototype transit-supportive districts around strategic nodes like Bukit Indah and Medini, leveraging its bus and planned rail network to anchor higher public transport usage.

From the application form to the inception phase

Iskandar Puteri (Johor, Malaysia) initially set out in its application form (January 2025) a broad smart-city agenda combining mobility optimisation, digitalisation of services, environmental conservation, and stronger citizen participation. Proposed solutions ranged from AI-driven traffic management and IoT adoption to green programmes like the Green Shield initiative and extensive public–private partnerships.

During the inception phase (July 2025), field visits, surveys, and multi-stakeholder workshops revealed mobility—especially in the Bukit Indah area—as the city's most pressing and cross-cutting challenge. Key gaps included heavy reliance on private cars, fragmented and low-ridership bus services, poor first/last-mile connectivity, limited universal design, and weak data integration and governance.

Figure 26 Bus station in Iskandar Puteri (Malaysia)

The ASUS Project – Phase II will support Iskandar Puteri in advancing integrated urban mobility and placemaking. The objective is to create a more liveable and transit-oriented city by improving first- and last-mile access, enhancing wayfinding, and linking public-realm investments with mobility nodes.

Next steps & planned support

The project team will support the city in developing a City Technical Proposal focusing on integrated, inclusive and future-ready urban mobility in Bukit Indah area (Iskandar Puteri), by developing an active mobility strategy to connect neighbourhoods, create a sense of place and facilitate long-term socio-economic resilience within the city and region. GEDSI considerations will be mainstreamed in the City Technical Proposal. To implement the City Technical Proposal, the city will seek funding allocations from the Iskandar Puteri City Council, although the city remains open to additional grant opportunities:

Activities planned in 2026:

- **Q1 2026**: Co-design workshop for the City Technical Proposal development
- **June 2026**: Completion of the City Technical Proposal

Potential funding sources

- Municipal budget
- Private-Public Partnership budget
- National Government Budget

Figure 27 end of the site visit in Iskandar Puteri with city officials

ASUS Working Group composition

City focal point	Mr Shahrulizam Abdul Rashid, Deputy Secretary, Iskandar Puteri City Council (MBIP)
Technical working group	• TPr Chew Lee Ting, Assistant Director – Project Manager • Ms Farah Izzati binti Muhamad Fuad, Team Lead – Smart City • Mr Mohamad Shahrul Izwan bin Azman, Engineer – Irrigation • Mr Mohammad Iman Arief bin Ayep, Engineer – Public Amenities • Mr Safwan bin Shaari, Team Lead – Sustainability and Urban Wellbeing • Mr Abdul Razak bin Seman, Architect • Ms Norafishyah binti Ahmad, Assistant Architect • Ms Nur Afiqah binti Sabri, Town Planner – Development Plan and Mobility

YANGON (MYANMAR) – SAFE PUBLIC SPACES

City Background

Yangon, Myanmar's largest and most economically dynamic city, lies along the Yangon River near the Ayeyarwady Delta. The metropolitan area spans approximately 600 km² and is home to over **7.3 million residents** (2024 Census), accounting for nearly one-third of Myanmar's total urban population. Once the nation's capital, Yangon remains its primary commercial and cultural hub, attracting internal migrants and international investment.

The city's rapid urbanisation over recent decades has reshaped its built environment, with farmland and wetlands converted into residential and industrial zones. Population density averages **716 persons/km²**, the highest in the country. However, this growth has intensified challenges such as housing shortages, traffic congestion, flooding, and loss of open space—now reduced to less than **0.5 m² of green area per resident**. These trends underscore the urgency of sustainable land use, public space revitalisation, and resilience-building to preserve Yangon's liveability and cultural identity.

Current resources and policies

In Yangon, a wide range of stakeholders influence the development and condition of public spaces, but in practice neighbourhood-level groups and community structures play an increasingly important role, especially in areas where formal oversight is limited or uneven.

The Yangon City Development Committee (YCDC) remains the body responsible for city administration and public service delivery through its 20 departments, including land-use regulation, infrastructure management, and the creation and maintenance of public spaces. Its Park Management Programme oversees parks, playgrounds, and recreational centres, and greening activities continue to enhance streetscapes and traffic islands. Heritage conservation efforts and disaster risk reduction initiatives-such as fire safety programmes and retrofitting for seismic resilience-remain part of the city's broader urban agenda. YCDC also implements the Yangon Zoning Regulation (introduced in 2014), Development Permit Rules (formalised in 2018), and environmental and public health regulations (derived from the City of Yangon Development Law of 1990) that set the formal framework for land use, density, construction, water supply, sanitation, and waste management.

However, community-level stakeholders are often the ones who sustain, adapt, and activate public spaces on a day-to-day basis, particularly in neighbourhoods where formal resources are constrained. Local groups commonly support park upkeep, organise clean-ups, and informally monitor safety issues. In many wards, residents play a key role in maintaining shared spaces, addressing small-scale improvements, and creating more inclusive and accessible environments through voluntary participation.

Based on available public information, other stakeholders also influence how public spaces evolve across Yangon's neighbourhoods:

- The Ministry of Natural Resources and Environmental Conservation (MONREC) provides national environmental policy oversight, including standards related to greening and climate resilience.
- The Ministry of Construction and the Department of Urban and Housing Development (DUHD) guide national urban development policies, housing programmes, and master planning.
- The Yangon Regional Government (YRG) shapes regional priorities and budget allocations.
- The Forest Department manages protected and forested areas within the Yangon region.

Alongside these formal structures, community-based organisations, ward-level groups, and local associations frequently contribute to maintaining and activating public spaces, ensuring they are safe and usable for residents. The private sector and development partners have also supported Yangon's urban initiatives through technical inputs, funding, or pilot projects.

While these stakeholders collectively shape the landscape of public spaces, overlapping mandates, limited budgets, and gaps in coordination have made equitable access and consistent maintenance challenging. This reinforces the importance of community-level stewardship and community initiative in sustaining safe and inclusive public spaces across Yangon.

Priority challenges & opportunities

Consultations during the inception phase identified several interlinked challenges:

- **Weak institutional coordination** and low community participation undermine effective urban planning.
- **Limited technical capacity** in landscape design, maintenance, and digital monitoring restricts the quality of green infrastructure.
- **Financial constraints** delay implementation and maintenance of public amenities.
- **Environmental degradation**, including flooding and heat stress, affects liveability, while **safety and security concerns** reduce the accessibility of public areas at night for vulnerable groups such as women, children, the elderly and persons with disabilities.

At the same time, Yangon presents promising opportunities for quick wins and scalable interventions:

- Strategic locations such as **riverbanks, ponds, and temple precincts** can host small-scale green public spaces.
- Collaboration with civil society organisations **(CSOs), academia, and private sector** can mobilise technical expertise and alternative financing.
- **Nature-based solutions**—rain gardens, permeable pavements, and tree corridors—can address drainage and climate adaptation while improving aesthetics and social interaction.

These opportunities can build confidence among local urban stakeholders and demonstrate the value of integrated, inclusive public space planning.

From the application form to the inception phase

The city's ASUS II application proposed redeveloping the green corridor along the Gyo Phyu Pipeline, converting an underutilised utility zone into a network of pedestrian paths, playgrounds, and community spaces. This initiative directly supports Yangon's Smart City Vision and Strategic Urban Development Plan 2040 (SUDP), which emphasises creating a "blue, green, and gold city."

During the inception workshop, this proposal evolved into a broader agenda for safe, inclusive, and climate-resilient public spaces, aligning with the ASUS sub-area on Personal Safety and Security. Stakeholders agreed that public space revitalisation offers both social and environmental benefits—enhancing wellbeing, fostering community cohesion, and mitigating flood risk through green infrastructure. The local technical proposal will therefore focus on planning and designing safe public spaces at the community level.

Next steps & planned support

In line with the UN engagement guidelines, the project team will support partners such as community representatives, CSOs, universities, and professional associations in developing a local technical proposal. Planned activities in 2025 and 2026 include:

- **November 2025:** Online training initiatives on inclusive urban design, climate-responsive landscaping, and digital monitoring of public spaces.
- **Q1 2026:** Technical support to refine the Diagnostic Report and the City Technical Proposal, including spatial analyses, conceptual designs, and pilot interventions design related to safe public spaces
- **Q2 2026:** Partnership facilitation between academic institutions, and private firms like Doh Eain or Think Maritime Design & Consultancy for co-implementation of demonstration projects.
- **May 2026:** Completion of the local technical proposal

Potential funding sources

International grants to implement green corridors and public-space projects.

ASUS Working Group composition

The Yangon City Consultation Workshop (16-17 September 2025) established a multi-stakeholder working group to guide the ASUS Project - Phase II implementation. It includes representatives from:

- **Local NGOs and CSOs** (3Zero House, Thant Myanmar, Young City Shaper Network)
- **Academic and professional institutions** (Yangon Technological University, Myanmar Architect Council, Myanmar Engineering Society)
- **Private sector and social enterprises** (Doh Eain, The Water Agency)
- **International organisations** (JICA, EU)

CEBU CITY (THE PHILIPPINES) – MOBILITY

City Background

Cebu City is the economic, cultural, and political centre of the Central Visayas, anchoring Metro Cebu with a population of nearly 1 million. The city covers 29,966 hectares, of which only 23% is urban, while the remaining 77% consists of upland and mountainous areas. This mix of flat coastal core and steep hinterlands shapes both opportunities and constraints in urban development. Cebu is historically known as the country's oldest city and today continues to evolve as a dynamic hub for commerce, BPO services, innovation, and education.

Current resources and policies

Cebu City's mobility ecosystem is complex, spanning a 1,025 km road network, 90 signalised intersections, and nine transport terminals. Recent initiatives demonstrate a shift toward sustainable and digital mobility: the Cebu Bus Rapid Transit (BRT) system is under development, complemented by the Local Public Transport Route Plan to update outdated jeepney routes. The city has also launched a bicycle lane network, with 90% of Phase 1 completed, and has established a digital Transportation Command Centre with 323 surveillance cameras. These initiatives are supported by policy instruments such as the Comprehensive Land-Use Plan (CLUP) 2023-2032 (approved in 2025), Local Public Transport Route Plan (, LPTRP) (approved in 2025), and Cebu's Traffic Code, although many of these frameworks require updating to match current realities.

Figure 28 Consultation with Cebu (The Philippines)

Priority challenges & opportunities

Cebu faces acute congestion driven by rapid motorisation, low-capacity jeepneys, and fragmented transport routes. Flash flooding frequently compounds mobility disruptions. At the same time, Cebu has strong opportunities to lead nationally: the existing command centre provides a ready platform for intelligent transport systems, while the BRT and bike lane expansion could transform commuter behaviour. The city's recognition of the need for hazard integration into transport management opens the door for more resilient and inclusive approaches.

Figure 29 Existing Data command center in Cebu (The Philippines)

From the application form to the inception phase

The city submitted an application identifying Mobility as its priority sub-area and proposing solutions to ease traffic congestion, improve transport, and address housing shortages for low-income and marginalised communities. Central to these solutions is the digitalisation of the transport system to support full pedestrianisation and an efficient public transport network. The inception phase provided an opportunity to refine the project's focus. Building on Cebu's existing digital command centre, the project will now prioritise upgrading the facility into an Intelligent Transportation System (ITS) hub. This upgrade will enable real-time traffic monitoring and response, integrate disaster risk management, pedestrian and non-motorised transport data, and leverage AI-driven analytics.

Next steps & planned support

The project team will support the city in developing a City Technical Proposal that details the upgrading of the command centre, including components related to feasibility and financial studies, capacity-building programs based on the identified issues, GEDSI analysis, and coordination with national and academic partners. Engagement with development finance institutions will be explored once the technical proposal is finalised to mobilise resources for hardware, software, and training. Parallel support will ensure the alignment of the command centre's future role with Cebu's Comprehensive Land Use Plan, Local Public Transport Plan, and with the new Bus Rapid Transit System:

- **Q4 2025**: completion of the city diagnostic report
- **February 2026**: co-design workshop for the development of the City Technical Proposal
- **June 2026**: completion of the City Technical Proposal

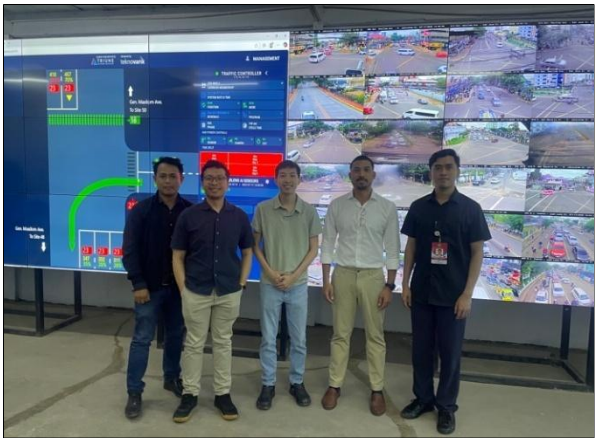

Figure 30 Assessment of the data command centre capabilities in Cebu (The Philippines)

Potential funding sources

- Municipal budget
- Grant Application from Korea Smart City Cooperation Fund

ASUS Working Group composition

City focal point	1. Hon. Nestor Archival, Mayor, Office of the City Mayor 2. Arch. Mrs. Ann Marie Cuizon, City Planning Officer, City Planning and Development Office, 3. Atty. Mr. Kent Jongoy, City Transportation Office 4. Ar. Mr. Jhomari Villarojo, City Planning and Development Office
Steering committee	• Representatives from Cebu City Mayor's Office (Mayor Mr. Nestor Archival), • Representatives from Cebu City Planning and Development Office (Arch. Mrs. Ann Marie Cuizon, EnP. Ms. Geraldine dela Cerna, Arch Mr. Jhomari Villarojo), • Representatives from Cebu City Transportation Office (Atty. Mr. Kent Jongoy, Jeffrey Lauderas, Mr. James Andrin), • Department of Transportation representatives (Ms. Haide

	Senajon, Mr. Jude Camponales, Mr. Micho Lauron) • Department of Human Settlements and Urban Development representative/s (Ms. Lara Quijano) • Department of the Interior and Local Government (ASCN representative) • Department of Economy, Planning, and Development (ACSI representative)

Figure 31 Traffic counting using existing CCTV infrastructure in Cebu (The Philippines)

CALOOCAN CITY (THE PHILIPPINES) – HOUSING

City Background

Caloocan City, with 1.66 million residents, is one of the most densely populated cities in Metro Manila. It occupies a fragmented territory divided between North and South Caloocan, giving it a unique urban structure. The city combines a thriving commercial and industrial base in the south with vast tracts of developable land in the north, positioning it as a key growth corridor. However, the city also faces acute housing pressures, with tens of thousands of informal settler families (ISFs) in need of secure, safe, and resilient shelter.

Figure 32 Site visit of a social housing project in North Caloocan (The Philippines)

The housing sector is central to Caloocan's urban agenda. National agencies such as Department of Human Settlements and Sustainable Urban Development and the National Housing Agency are actively implementing large-scale projects under the government's 4PH (Pambansang Pabahay para sa Pilipino) program, with over 10,000 housing units under construction. Complementary projects include socialised housing, SHFC-supported led by National Housing Agency, community mortgage schemes supported by the Social Housing Finance Corporation (SHFC), and city-initiated tenurial upgrading efforts. The city's Local Shelter Plan (2026–2035) is being updated, and numerous council resolutions demonstrate political support for addressing Informal Settlers Families (ISF) challenges. These initiatives highlight strong alignment between local and national priorities, although financial and institutional gaps remain.

Priority challenges & opportunities

Caloocan has recorded 58,657 ISFs, with the majority still awaiting relocation. Many are located in waterways, road-right-of-way, and danger-prone areas, exposing families to high levels of vulnerability. Relocation efforts remain uneven, with some categories progressing faster than others. At the same time, the city has opportunities to scale up innovative solutions: partnerships with national housing programs are well established, and tenurial upgrading projects offer a pathway to more secure housing. The city has also expressed openness to partnerships on green and climate-resilient housing solutions, which can improve both livability and long-term resilience.

From the application form to the inception phase

The city application form identified Housing as its top priority, proposing broad solutions, such as updating land-use and shelter plans, strengthening social housing, partnering with private and international funders, and piloting smart housing innovations such as using prefabricated materials, modular housing and eco-friendly designs. During the inception phase, field visits and stakeholders' consultations revealed key gaps – such as affordability issues, drainage and sewage problems, and the need for climate resilience in existing housing projects. This led to a sharper focus, supporting green housing retrofits within current social housing programs, integrating nature-based solutions and climate-resilient designs.

Figure 33 Site visit of a social housing project still being constructed in South Caloocan (The Philippines)

Next steps & planned support

The project team will support Caloocan City in developing a City Technical Proposal focused on tenure upgrading and resilient housing. Planned support includes exploring grant opportunities with the "Partnership for Sustainable Building in Southeast Asia" (PEEB ASEAN), funded by the French Development Agency, to seek an additional support for energy-efficient and resilient building.

Figure 34 Mapping exercise with community representatives, local NGOs and city officials in Caloocan (The Philippines)

Under the City Technical Proposal, capacity-building activities will be proposed to the Housing and Resettlement Office and other local agencies to enhance governance, financing, and project design capacities:

- **Q4 2025**: completion of the city diagnostic report
- **February 2026**: co-design workshop for the development of the City Technical Proposal
- **June 2026**: completion of the City Technical Proposal

Potential funding sources

- Grant Application from Partnership for Energy Efficiency in Buildings-ASEAN from Agence Française de Développement (AFD)

Figure 35 Common areas of a social housing project in Caloocan (The Philippines)

ASUS Working Group composition

City focal point	1. EnP Aurora C. Ciego, DPA (City Administrator) 2. Ms. Josephine Dela Cruz, City Planning Officer (Officer in charge), City Planning and Development Office, 3. Ms. Ma. Relizah Cruz (Housing and Resettlement Office representative) 4. Mr. Erick Castillo (Housing and Resettlement Office representative) 5. Dr. (Mr.) Rhod Dantay, Head of Housing and Resettlement Office
Steering committee	• Representatives from Caloocan City Administrator's Office (EnP. Ms. Aurora Ciego, Mr. Al Fajardo),

	• Representatives from Caloocan City Planning and Development Office, (EnP. Ms. Josephine Dela Cruz, Mr. Joselito Fausto, Mr. Reymark Javillonar), • Representatives from Caloocan City Housing and Resettlement Office (Ms. Relizah Cruz, Mr. Roderick Castillo, Ms. Jami Sabilano, Mr. Benedict Simon, Ms. Lei Sarmiento) • Department of Human Settlements and Urban Development (Mr. John Soriano) • Socialized Housing Finance Corporation (Mr. Joshua Delvalo, Mr. Ryan Ariola) • Department of Interior and Local Government (ASCN representative) • Department of Economy, Planning, and Development (ACSI representative)

NAKHON SI THAMMARAT (THAILAND) – SOLID WASTE MANAGEMENT

City Background

Nakhon Si Thammarat, a historic city on Thailand's east coast, is home to roughly 100,000 residents. Built on a linear sand ridge, the city has a dense urban footprint and a long history as a regional hub. Seasonal monsoons and tropical storms are a constant feature of local life and have a direct impact on the city's waste governance systems, particularly landfill operations, which remain the cornerstone of municipal solid waste management.

Figure 36 Site visit of the Landfill facility in Nakhon Si Thammarat (Thailand)

Current resources and policies

The city's primary landfill site at Thung Talad has accumulated more than 2 million tonnes of waste, partly due to intake from surrounding districts. Court rulings have now restricted access to three entities, but the site remains overstretched. The municipality generates between 110 and 248 tonnes of waste per day, depending on the source of data, and is shifting policy toward source segregation, recycling, and circular economy measures. These priorities are embedded in the Local Development Plan for 2023–2027, which calls for greater emphasis on recovery and reduced reliance on landfill.

Priority challenges & opportunities

The challenges are clear: limited landfill space, leachate and slope stability risks during the rainy season, and odour and fire hazards during the dry season. Fee collection and behaviour change remain additional hurdles. Yet the city also has opportunities. The policy direction has shifted decisively away from landfill-centric disposal, creating room to professionalise collection, sorting, and materials recovery. Market and school networks, as well as private haulers, can play a greater role in improving segregation and recovery if supported by clear municipal leadership.

Figure 37 Inception Workshop in Nakhon Si Thammarat (Thailand)

From the application form to the inception phase

The application form of Nakhon Si Thammarat (January 2025) prioritised water, waste and sanitation, aiming to create a sustainable solid waste management system through Reduce, Recycle, Reuse (3R) campaigns, waste segregation, informal sector engagement, and long-term engagement such as an Refuse-Derived Fuel plant and a waste-to-energy facility.

During the inception phase (May 2025), fieldwork and consultations revealed that the city's most urgent issue is the Thung Talad landfill, which holds 2 million tons of unmanaged waste and lacks leachate treatment, causing serious environmental, legal and social risks. The project will therefore support the city in focusing on downstream waste management, centring on rehabilitating and optimising the landfill with the Fukuoka semi-aerobic method, fulfilling court-mandated requirements, building institutional and data-monitoring capacity, and engaging communities and informal waste pickers.

Next steps & planned support

The project team will support the city in developing a City Technical Proposal focusing on enhancing downstream waste management with the improvement of operational efficiency, semi-aerobic landfill rehabilitation and strategic reduction of legacy waste stockpiles. As part of the City Technical Proposal, capacity-building activities will be included to help municipal staff improve their planning, monitoring and management of landfill operations using data-driven and regulatory compliant approaches. The project is also supporting the city in fulfilling its legal obligations in developing a City Technical Proposal demonstrating technical compliance, environmental risk reduction and responsible landfill operations. Through this approach, the ASUS Project – Phase II aims to reduce the environmental risks associated with landfill operations while building a more circular and resilient waste management system:

- Q4 2025: completion of the city diagnostic report
- February 2026: co-design workshop for the development of the City Technical Proposal
- June 2026: completion of the City Technical Proposal

Figure 38 Mapping exercise with community representatives in Nakhon Si Thammarat (Thailand)

Potential funding sources

National budget

ASUS Working Group composition

City focal point	Mr. Peemmawat Na Pattalung, Plan and Policy Analyst, Nakhon Si Thammarat City Municipality
Steering committee	• Mr. Chatchai Arunsuwankorn, Deputy Mayor of Nakhon Si Thammarat City Municiaplity • Mr. Phongsak Srirat, Division of Public Health and Environment Division • Mr. Ramet Paiboonsombat, Office of Public Works • Mr. Sunchai Chatretup, Deputy Clark, Division of Strategy and Budget

CHIANG MAI (THAILAND) – SAFETY & SECURITY

City Background

Chiang Mai, the largest province in northern Thailand, is home to nearly 1.8 million residents, with urbanisation concentrated in the valley floor around Mueang Chiang Mai. More than 70% of the province remains mountainous or forested, shaping both constraints and opportunities for urban development. Chiang Mai is globally recognised for its cultural and heritage significance—it is a UNESCO Creative City for Crafts and Folk Art and is listed on UNESCO's tentative World Heritage list. These assets, together with its reputation as a tourism and digital-nomad hub, have contributed to steady population growth and an increasingly complex urban economy.

Figure 39 Site visit of the data command centre in Chiang Mai (Thailand)

Current resources and policies

Urban land accounts for less than 4% of the province, placing significant pressure on the historic core and peri-urban settlements. In recent years, the city and province have invested in various digital and operational systems to manage tourism flows, traffic, and environmental services. However, datasets are fragmented across agencies, and operational coordination is limited. Stakeholders emphasised the urgent need for an integrated approach to city information management, with interoperable systems that support day-to-day operations in mobility, safety, and environmental monitoring.

The project aligns with Chiang Mai's "Livable City" vision under the Provincial Development Plan (2023–2027) and the Draft Smart City Development Strategy (2025–2027), establishing a clear provincial policy baseline complemented by national frameworks including the 20-Year National Strategy (2018–2037) and the Personal Data Protection Act (2019).

Priority challenges & opportunities

The main challenge lies in fragmented and incomplete data systems, which hinder evidence-based planning and incident management. Rising visitor and resident flows have made gaps in data governance more visible, particularly in areas such as environmental monitoring and hazard response. At the same time, Chiang Mai has significant opportunities: provincial authorities are committed to modernising digital infrastructure, and the city's status as a cultural and tourism hub creates strong incentives to develop a holistic urban data platform that can enhance safety, planning, and public service delivery.

Figure 40 Mapping exercise with city officials in Chiang Mai (Thailand)

From the application form to the inception phase

Chiang Mai's application form (January 2025) highlighted Safety and Security as its key priority, aiming to create a safer and smarter urban environment for residents and visitors. The city proposed integrating its 1,000+ CCTV cameras into a unified network, expanding coverage in economic zones, and using AI-driven analytics, emergency response platforms, and smart infrastructure (e.g., adaptive lighting, real-time traffic signage) to improve monitoring and response.

During the inception phase (May 2025), consultations and site visits showed that Chiang Mai already has strong digital infrastructure—over 2,700 CCTV cameras and a robust fibre network—but faces fragmentation across agencies (provincial government, municipality, Tourist Police, Provincial Police Region 5) and lacks real-time integrated data management.

The project will therefore focus on developing an integrated city data platform led by the Chiang Mai Provincial Administrative Organisation (PAO), with support from DEPA and UN-Habitat. Key actions in the City Technical Proposal will include the creation of a provincial working group, development of protocols for data sharing and privacy, piloting an AI-based CCTV analytics, and building capacity for smart urban management. GEDSI considerations will be mainstreamed in the in-depth analysis of the challenges and opportunities in the diagnostic report and proposed GEDSI interventions will be integrated in the City Technical Proposal.

Next steps & planned support

The project team will support the city in developing a City Technical Proposal (CTP) that sets out the scope of the city data platform, prioritises use cases that can leverage CCTV infrastructure for planning purposes related to Safety and Security (incident reporting, digital twin program), and establish a working group for joint planning related to smart urban management. In parallel of the CTP, the ASUS Project – Phase II is supporting the city in identifying relevant grants and funding opportunities. Under the City Technical Proposal, capacity-building activities will be proposed to strengthen data stewardship, improve agency coordination, and create operational playbooks for real-time incident response:

- **Q4 2025**: completion of the city diagnostic report
- **December 2025**: co-design workshop for the development of the City Technical Proposal
- **June 2026**: completion of the City Technical Proposal

Figure 41 Showcase of existing CCTV capabilities from a local university in Chiang Mai (Thailand)

Potential funding sources

- National budget
- Grant Application from Korea Smart City Application Fund
- Scoping of additional funding sources is an ongoing process

ASUS Working Group composition

City focal point	Mr. Pradya Komanee, Manager, Upper Northern Provincial Cluster Office Digital Economy Promotion Agency (depa)
Steering committee	• Mr. Siwa Tamikanont, Vice Governor of Chiang Mai Province Mr. Wutthiphong Srisin, Chiang Mai Provincial Administrative Organisation • Pol.Maj. Prapan Chotipattananon (MR.), Provincial Police Region 5 • Pol. Lt. Col. Makara Srisakolphisut (MR.), Chiang Mai Tourist Police • Mr. Pradya Komanee, depa • Asst. Prof. Dr. Suepphong Chernbumroong (MR.), Chiang Mai University

BAC GIANG (VIETNAM) – WATER, WASTE & SANITATION

City Background

Bac Giang Ward—formerly Bac Giang City—serves as the new administrative and political centre of the merged Bac Ninh Province, located about 50 km from Hanoi and Noi Bai Airport and 130 km from Hai Phong Port. The ward occupies 258 km² and hosts approximately 371,000 inhabitants across 31 wards and communes.

As one of Vietnam's fastest-growing localities, Bac Giang recorded GRDP growth rates of 14–19 per cent in recent years. Its strategic position along the Hanoi–Lang Son and Hanoi–Hai Phong economic corridors strengthen its logistics and industrial competitiveness. The 2025 administrative merger and new Grade I urban designation mark a decisive transition toward becoming a smart, green, and high-potential urban hub in the Northern Midlands region.

Current resources and policies

Bac Giang's institutional capacity has recently expanded through the creation of a ₫700 billion Inter-Agency Administrative Centre and a Smart City Operations Centre connecting all wards via an integrated e-governance platform. These investments have digitised city management and improved coordination between departments.

Physically, the city is supported by a strong transport and logistics network, including the Bac Giang–Lang Son Expressway, the Bac Giang–Ha Long Expressway, and new bridges improving regional connectivity.

Policy direction is framed by national directives on smart city development and climate-resilient growth, with the city's Urban Development Program to 2045 prioritising compact expansion, eco-urban zones, and renewable energy integration.

Ongoing interventions include:

- Industrial Park upgrading and logistics corridor expansion.
- Ecotourism and mountain resort development for economic diversification.
- Waste-to-energy plant construction in Da Mai Ward (750 tons/day) and Nham Bien Plant for solid waste conversion to energy

Priority challenges & opportunities

Flooding and poor drainage emerged as Bac Giang's core urban challenges, compounded by outdated infrastructure, fragmented drainage networks, and limited maintenance budgets. Rapid urban expansion and rising impermeable surfaces have worsened runoff and local flooding.

However, opportunities exist to integrate smart drainage systems, climate-adaptive infrastructure, and digital flood monitoring tools into future urban planning.

The city's flat terrain and central location make it suitable for compact Transit-Oriented Development (TOD) and integrated green-blue corridors, which could be supported through international financing and public-private partnerships.

From the application form to the inception phase

The city's application identified waste management, drainage, and digital governance as top priorities. During the inception phase, these priorities were refined into a targeted technical assistance package focusing on integrated flood and drainage management—linking flood diversion corridors with smart city operations and spatial planning.

Next steps & planned support

- Supporting the city in developing City Diagnostic Report and City Technical Proposal to mobilise external funding

- Exploring quick-win interventions, such as rainwater parks and drainage monitoring pilots

Activities planned in 2026:

- **January 2026**: completion of the city diagnostic report
- **February 2026**: co-design workshop for the development of the City Technical Proposal
- **June 2026**: completion of the City Technical Proposal

Potential funding sources

Bac Giang intends to mobilise state budget and Official Development Assistance resources.

ASUS Working Group composition

A multi-stakeholder working group has been established, comprising:

1. Mr. Nguyen Duy Hung, Vice Chairman of Bac Giang People's Committee
2. Ms. Ha Thi Hau Nga, Deputy Chief of Bac Giang Office of People's Committee
3. Mr. Trieu Tien Dat, Deputy Head of Economics, Infrastructure and Urban Affairs Department, Bac Giang People's Committee
4. Mr. Le Van Son, Deputy Director, Bac Giang Wastewater Treatment Center
5. Ms. Nguyen Thi Thuy Ninh, Executive Committee Member, Bac Giang Women's Union
6. Mr. Dinh Tuan Hai, Researcher, University of Civil Engineering
7. Mr. Nguyen Cong Thanh, Individual Senior Expert on Water Supply and Sanitation
8. Mr. Le Dinh Nam, Group Leader of Ngo Quyen Neighborhood
9. Mr. Nguyen Minh Giang, Group Leaderof Tho Xuong Neighborhood
10. Mr. Tran Duc Minh, Group Leader of Minh Khai Neighborhood
11. Mr. Bui Hong Quang, Group Leader of Song Mai Neighborhood
12. Mr. Vu Duc Phuc, Group Leader of Tran Phu Neighborhood
13. Mr. Tran Trung Hieu, Group Leader of Da Mai Neighborhood
14. Mr. Dinh Trong Ha, Group Leader of Hoang Van Thu Neighborhood

In Bac Giang ward, women's leadership reflects stronger voices with high-level position's participation.

HUE (VIETNAM) – URBAN RESILIENCE

City Background

Hue City, officially designated a centrally governed city in January 2025, covers 4,947 km² with a population of 1.24 million. Situated between Hanoi and Ho Chi Minh City along Vietnam's Central Economic Corridor, Hue is a key node for culture, tourism, and health services. The city is home to eight UNESCO World Heritage Sites and the Tam Giang – Cầu Hai lagoon, the largest in Southeast Asia. Hue's heritage-based urban identity and ecological assets make it both a symbol of Vietnam's cultural continuity and a testbed for climate-resilient urban development.

Current resources and policies

Hue possesses strong institutional and financial resources as a nationally prioritised urban centre. Policies such as Resolution No. 54-NQ/TW (2019) and Decision 108/QĐ-TTg (2024) define Hue as a "heritage, ecological, and intelligent city." The city's development is anchored in the Urban Development Program to 2045 and the Provincial Master Plan 2021–2030, both emphasising urban resilience, waste management, and heritage preservation.

Key ongoing projects include:

- The Hue City Water Environment Improvement Project (ADB-financed), constructing two wastewater plants with a combined capacity of 60,000 m³/day;
- The Phu Son Waste-to-Energy Plant (600 tons/day), producing 93 million kWh of green electricity annually;
- Climate-adaptation initiatives funded by the Green Climate Fund and Luxembourg Development Cooperation, supporting resilient finance and coastal forest restoration

Priority challenges & opportunities

Hue's rapid urbanisation has intensified flooding and transport congestion in core districts such as Thuan Hoa and Phu Xuan. Drainage coverage reaches only 60–70 per cent of urban land, and combined sewer systems are overloaded. Climate change exacerbates risks of inundation and coastal erosion.

Nevertheless, Hue holds distinct opportunities:

- Its status as a heritage city enables international support for resilient infrastructure integrated with heritage preservation.
- Existing climate projects provide a strong foundation for cross-sector coordination.
- Large-scale data systems such as the Hue-S app can be expanded for urban monitoring and public feedback.

From the application form to the inception phase

The application initially prioritised urban flood management, wastewater treatment, and housing. During the inception workshop, these were refined into a single integrated focus on urban resilience in flood-prone core districts, linking water management, mobility, and heritage preservation. This focus reflects Hue's strategic goal of operationalising the ASUS framework through evidence-based planning and pilot interventions that demonstrate flood risk reduction and liveability improvement.

Next steps & planned support

- Support the city in developing a City Diagnostic Report and City Technical Proposal focusing on flood and drainage management;
- Assessment of adequate funding opportunities for the implementation of the City Technical Proposal from international sources

Activities planned in 2026:

- **January 2026**: completion of the city diagnostic report
- **February 2026**: co-design workshop for the development of the City Technical Proposal
- **June 2026**: completion of the City Technical Proposal

Potential funding sources

Hue intends to mobilise state budget and Official Development Assistance resources.

ASUS Working Group composition

The Hue ASUS Working Group includes:

1. Mr. Le Chi Kien, Deputy Chief of Office, Hue People's Committee
2. Mr. Truong Nguyen Thien Nhan, Deputy Director General, Departments of Construction
3. Mr. Ngo Quang Thinh, Deputy Head of Urban Development & Technical Infrastructure Division, Departments of Construction
4. Mr. Dong Huu Huy Hoang, Head of Architecture & Planning Division, Departments of Construction
5. Mr. Le Van Binh, Deputy Head of Irrigation and Climate Change Agency, Department of Agriculture and Environment
6. Mr. Cung Trong Cuong, Director General, Hue Institute for Development Studies under Hue People's Committee
7. Ms. Dang Thi Kieu Nhi, Official of Urban Development Research Department, Hue Institute for Development Studies
8. Mr. Tran Anh Tuan, Professor, Hue University of Sciences
9. Mr. Le Vinh Thang, Deputy General Manager, Hue Urban Environment and Public Works JSC
10. Ms. Nguyễn Thị Tâm An, Official of Thuan Hoa Ward People's Committee
11. Mr. Nguyen Van Thang, Official of Phu Xuan Ward People's Committee
12. Ms. Le Thi Hoai Thuong, Official of Phu Xuan Ward People's Committee

Women's leadership is reflected better in the grassroot level e.g ward and commune areas where the project conducted consultations.